

WAKEFIELD, VA, USA

WMO: 724019

Lat: 36.983N

Lon: 77.001W

Elev: 111

StdP: 14.64

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 93773

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.6	21.7	0.4	5.6	25.1	5.2	7.2	28.4	16.6	53.9	15.0	50.2	2.1	350	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	94.1	76.4	91.7	75.6	89.8	74.9	79.0	89.6	78.0	88.0	77.0	86.2	5.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.1	136.8	83.2	75.1	132.2	82.5	74.2	128.0	81.5	42.5	89.8	41.4	88.3	40.3	86.3	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.8	11.9	10.5	8.2	98.3	8.4	3.0	2.1	100.4	-2.7	102.2	-7.4	103.8	-13.5	106.0		
(5)			6.9	81.6	7.9	1.3	1.2	82.5	-3.4	83.3	-7.9	84.0	-13.6	85.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.8	39.7	42.5	49.0	59.2	67.6	75.3	78.7	77.6	71.7	60.7	50.3	43.5		
	DBStd	16.05	10.79	9.72	9.97	8.92	7.55	5.77	4.73	4.39	6.21	8.36	8.72	9.59		
	HDD50	1133	352	251	147	18	0	0	0	0	0	11	102	252		
	HDD65	3523	785	630	504	219	64	3	0	0	17	186	447	668		
	CDD50	4693	33	41	116	295	546	760	890	857	651	342	111	51		
	CDD65	1608	1	1	7	45	146	313	425	392	218	52	6	2		
	CDH74	14777	2	8	77	468	1289	3014	4259	3560	1678	385	31	6		
	CDH80	5818	0	0	8	122	418	1250	1897	1478	554	87	4	0		
(14) Wind	WSAvg	3.6	4.1	4.4	4.7	4.6	3.7	3.3	3.1	2.5	2.7	3.0	3.4	3.5		
(15) Precipitation	PrecAvg	45.8	3.4	2.6	4.3	3.2	3.6	4.3	5.1	4.6	4.8	3.3	3.0	3.5		
	PrecMax	61.0	6.4	5.7	7.4	6.7	7.0	9.1	7.9	10.8	19.9	6.1	7.9	6.9		
	PrecMin	34.2	1.1	0.8	0.7	1.2	1.3	1.9	2.4	1.5	0.4	0.0	0.5	1.0		
	PrecStd	6.9	1.4	1.2	1.7	1.3	1.4	2.0	1.7	2.3	4.1	1.7	1.9	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.7	74.9	81.4	88.7	91.3	96.8	98.4	95.9	92.5	86.6	78.6	74.0		
		MCWB	62.1	63.0	62.6	67.3	72.6	75.7	77.5	77.3	73.8	71.8	67.1	65.1		
	2%	DB	66.4	69.8	75.9	83.5	87.9	92.7	94.5	92.8	89.2	82.1	73.1	69.0		
		MCWB	59.7	59.0	60.3	66.1	71.1	75.1	77.3	76.2	73.7	69.8	63.0	63.1		
	5%	DB	62.7	64.6	71.7	79.6	84.6	90.1	91.8	90.4	86.2	78.4	69.8	64.4		
		MCWB	56.6	56.1	58.3	63.8	70.2	74.7	76.6	75.6	72.6	67.8	61.9	59.7		
	10%	DB	57.3	60.7	66.9	75.2	81.6	87.5	89.6	87.9	82.9	74.6	65.5	60.5		
		MCWB	51.0	52.3	56.7	62.8	68.2	73.6	75.6	74.7	71.6	65.9	59.3	55.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.5	65.8	65.9	71.3	75.3	78.9	80.7	80.5	77.9	76.1	69.7	67.9		
		MCDB	68.8	71.8	75.1	81.1	85.9	90.6	91.6	89.8	86.8	80.7	73.8	72.2		
	2%	WB	61.6	62.1	63.7	68.8	73.4	77.4	79.3	78.8	76.4	73.1	66.3	63.9		
		MCDB	65.8	67.3	71.1	79.0	83.3	88.2	90.8	88.2	84.2	79.4	71.4	67.5		
	5%	WB	57.4	58.2	61.5	66.6	72.0	76.3	78.2	77.8	75.4	70.4	63.3	60.6		
		MCDB	61.6	64.2	70.4	75.9	81.0	86.4	88.8	87.0	82.2	75.9	67.0	64.3		
	10%	WB	52.3	52.8	58.4	64.3	70.5	75.2	77.1	76.6	74.0	67.7	60.1	56.0		
		MCDB	56.6	58.8	65.5	72.6	78.7	84.4	86.7	84.8	79.7	72.9	65.2	59.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.5	21.0	22.3	23.2	21.0	20.0	19.2	18.9	19.2	21.7	22.0	19.6		
		MCDBR	24.0	26.6	27.1	26.2	23.7	22.5	21.7	21.3	22.3	23.4	24.9	21.8		
	5% WB	MCWBR	17.9	18.3	14.8	12.0	10.7	8.9	8.1	7.9	9.5	12.3	15.6	17.0		
		MCWBR	21.9	24.8	23.0	22.0	20.0	19.6	19.1	18.0	17.1	18.5	19.5	20.4		
(40) Clear-Sky Solar Irradiance	taub		0.302	0.316	0.345	0.390	0.432	0.488	0.513	0.509	0.405	0.354	0.321	0.306		
	taud		2.531	2.489	2.437	2.323	2.233	2.127	2.075	2.072	2.355	2.491	2.537	2.556		
	Ebn at Noon		284	292	292	283	271	255	247	245	268	275	274	275		
	Edh at Noon		25	29	34	40	44	49	52	51	36	29	24	23		
(44) All-Sky Solar Radiation	RadAvg		753	989	1307	1659	1867	2013	1914	1728	1407	1118	845	666		
	RadStd		61	111	105	127	161	132	94	92	126	135	68	55		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+301	N/A
(47) Regional (67 neighbors)		+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+301	+193

Nomenclature: See separate page